

Title: Portable Glaucoma Screening Device Using Retinal Image Analysis

Abstract:

Glaucoma is a leading cause of irreversible blindness worldwide, caused by progressive optic nerve damage. Early detection is crucial, but conventional diagnostic methods are often expensive and clinic-based, limiting access in remote areas. This project presents a portable, cost-effective device for preliminary glaucoma screening using retinal image analysis.

The system utilizes an ESP32-CAM module with an OV2640 camera and a +20D Volk lens to capture fundus images, supported by LED illumination. Images are processed using computer vision algorithms to segment the optic disc and cup, and calculate the Cup-to-Disc Ratio (CDR), a key indicator of glaucoma.

Based on the CDR, the device classifies results into “**Normal**” ($\text{CDR} < 0.5$) or “**Glaucoma Suspect**” ($\text{CDR} \geq 0.5$) and displays the outcome on an integrated LCD screen. This standalone system eliminates the need for smartphones or external computers, making it suitable for **screening in remote and low-resource settings**.